

Relation Algebra

§10. Freyd's Power allegories

(10a)

A **power allegory** is a division allegory with one unary operation on arrows, $\exists$ **epsilon**, subject to

$$\begin{aligned} \exists_R \square &= R\square, \quad \exists_R = \exists_R \square \\ \mathbb{1} &\sqsubseteq (R/\exists_R)(\exists_R/R) & \exists_R \text{ is } \mathbf{thick} \\ \frac{\exists_R}{\exists_R} &= \mathbb{1} & \exists_R \text{ is } \mathbf{straight} \end{aligned}$$

$R\square$ is R 's target, an identity arrow. For $R : A \rightarrow B$ write $\exists : EB \rightarrow B$, dropping the subscript.

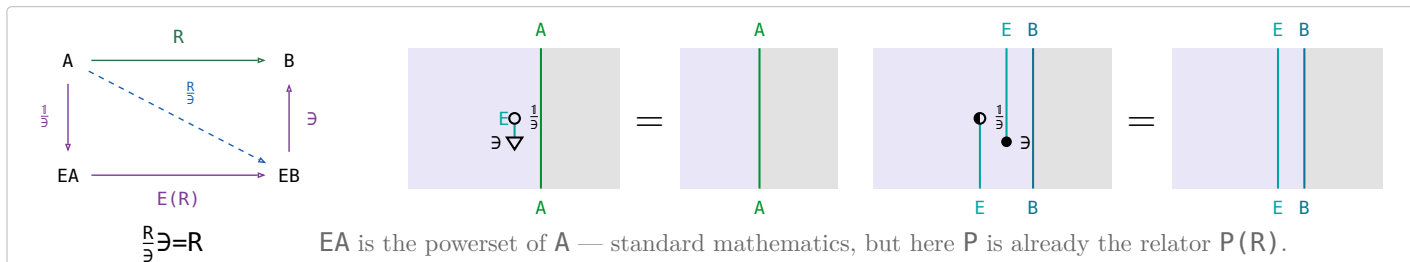
$$\epsilon \triangleq \exists^\circ : A \rightarrow EA$$

(10b)

law	the reading
1 $\exists_R \square = R\square, \quad \exists_R = \exists_R \square$	One $\exists$ per object, not per arrow.
2 $\exists$ is thick	Comprehension: every x has a set of exactly the people x admires.
3 $\frac{R}{\exists} : A \rightarrow EB$, for $R : A \rightarrow B$	convert a relation to a function. $\mathbf{a} \frac{R}{\exists} = \{b \mid a R b\}$
4 $\mathbf{Map}(\frac{R}{\exists})$	
5 $\frac{R}{\exists} \exists = R$	reading the chosen set back through $\exists$ returns the relation.
6 $\mathbf{Simple}(F) \Rightarrow F \sqsubseteq \frac{F\exists}{\exists}$	A partial choice of sets is inside the total one.
7 the singleton map is monic: $\frac{1}{\exists}$ $(\frac{1}{\exists})^\circ \sqsubseteq \mathbb{1}$ two points with the same one-person set are the same point.	The one-person set.
8 $\frac{\exists}{\exists} = \mathbb{1}$	Make the set of a set, then read it back one level down.
9 fusion: $\mathbf{Map}(f) \Rightarrow \frac{fR}{\exists} = f \frac{R}{\exists}$	Naturality of the unit, $\mathbf{Map}(f) \Rightarrow f \frac{1}{\exists} = \frac{1}{\exists} E(f)$. renaming a point and then taking its one-person set is taking the set and renaming inside it.
10 $\mathbf{Map}(f) \wedge \mathbb{1} \square = \exists \square \Rightarrow \frac{f}{\exists} = f \frac{1}{\exists}$	Rename first or take singletons first — the fusion row above at $R=1$.
11 $R\square = \exists \square \Rightarrow R/_s S = \frac{R}{\exists} (\frac{S}{\exists})^\circ$	x and y match when R sends x and S sends y to the same set.
12 $E(R) \triangleq \frac{\exists R}{\exists}$	$E(R) : EA \rightarrow EB$, image of a set of A
13 $\frac{R}{\exists} = \frac{1}{\exists} E(R)$	$\frac{1}{\exists} : x \mapsto \{x\}$
14 $\frac{S}{\exists} E(R) = \frac{SR}{\exists}$	absorption — the monad's composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{SR}{\exists}$, §Section 10.3
15 $\exists \triangleq \exists / \exists$	$xs \supseteq ys \iff \forall a. ys \exists a \rightarrow xs \exists a$
16 $\sqsubseteq = \supseteq^\circ$	$xs \sqsubseteq ys \iff ys \supseteq xs$

§10.1. $i \vdash E$ Power Allegory defined as adjunction

(10.1a)



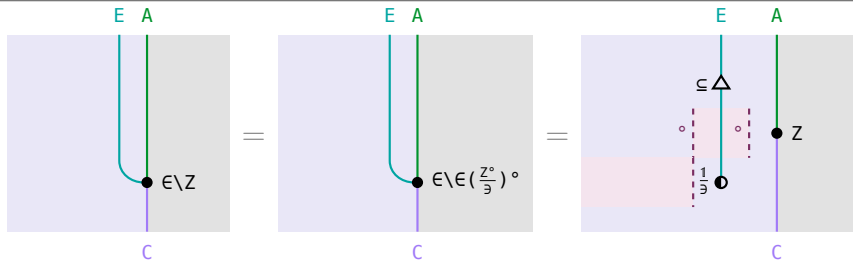
i is the inclusion $\mathbf{Map}(\mathcal{A}) \rightarrow \mathcal{A}$, doing nothing, so E is both the functor and the monad iE .

§10.2. $\epsilon \setminus$ as a composite

$$\epsilon \setminus Z = \subseteq \left(\frac{Z^\circ}{\exists} \right)^\circ$$

(10.2a)

$\mathbf{x}\mathbf{s}$ is related to $\mathbf{c}$ by $\epsilon \setminus Z$ exactly when $\mathbf{x}\mathbf{s} \subseteq \Lambda(Z^\circ)(\mathbf{c})$, the set of every $\mathbf{a}$ with $\mathbf{a} Z \mathbf{c}$



$$EA \boxed{\epsilon \setminus Z} C$$

$$= EA \boxed{\epsilon \in \left(\frac{Z^\circ}{\exists} \right)^\circ} C$$

$$= EA \boxed{\subseteq} \boxed{E(Z^\circ)} EC \boxed{\frac{1}{\exists}} C$$

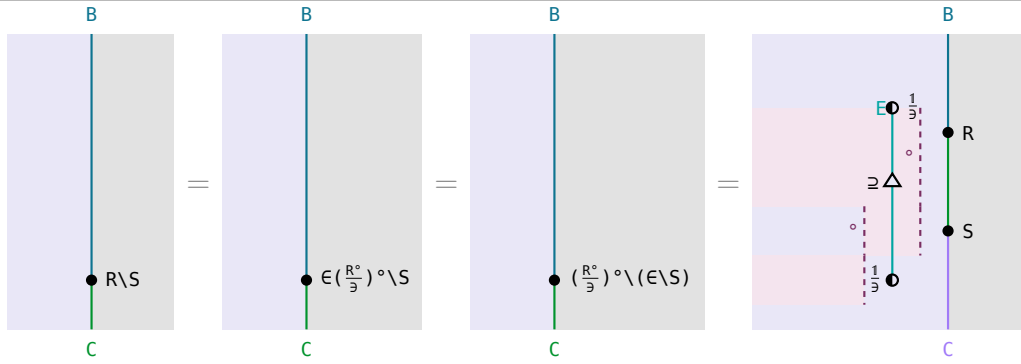
$Z = \epsilon \Lambda(Z^\circ)^\circ$: the converse of $\Lambda(Z^\circ) \ni Z^\circ$ — (10b)

$X \setminus (Yf^\circ) = (X \setminus Y) f^\circ$ for a map f (not a tabulated row), at the map $f \triangleq \Lambda(Z^\circ)$; $\subseteq \triangleq \epsilon \setminus \epsilon$

$$R \setminus S = \frac{R^\circ}{\exists} \subseteq \left(\frac{S^\circ}{\exists} \right)^\circ$$

(10.2b)

$\mathbf{b}$ is related to $\mathbf{c}$ by $R \setminus S$ exactly when $\Lambda(R^\circ)(\mathbf{b}) \subseteq \Lambda(S^\circ)(\mathbf{c})$, the $\mathbf{R}$ -preimage of $\mathbf{b}$ inside the $\mathbf{S}$ -preimage of $\mathbf{c}$



$$B \boxed{R \setminus S} C$$

$$= B \boxed{\epsilon \in \left(\frac{R^\circ}{\exists} \right)^\circ \setminus S} C$$

$$= B \boxed{\left(\frac{R^\circ}{\exists} \right)^\circ \setminus (\epsilon \setminus S)} C$$

$$= B \boxed{\frac{1}{\exists}} EB \boxed{E(R^\circ)} \boxed{\subseteq} EA \boxed{E(S^\circ)} EC \boxed{\frac{1}{\exists}} C$$

$R = \epsilon \Lambda(R^\circ)^\circ$: the converse of $\Lambda(R^\circ) \ni R^\circ$ — (10b)

$(XY) \setminus S = Y \setminus (X \setminus S)$

$f^\circ \setminus X = fX$ for the map $f \triangleq \Lambda(R^\circ)$, and $\epsilon \setminus S = \subseteq \Lambda(S^\circ)^\circ$ — (10.2a)

§10.3. $\mathcal{A} \cong \mathbf{Kleisli}(E)$

(10.3a)

$E(R) \triangleq \frac{\exists R}{\exists}$, $\frac{1}{\exists} : A \rightarrow EA$, $\text{union} = E(\exists) : E(EA) \rightarrow EA$

$f \diamond g \triangleq f E(g) \text{union}$, for $f : A \rightarrow EB$ and $g : B \rightarrow EC$

the monad is on $\mathbf{Map}(\mathcal{A})$, not on the allegory: E is a relator on all relations, but $\frac{1}{\exists}$, union and $f E(g) \text{union}$ are maps, and the Kleisli construction happens where they live.

(10.3b)

$$\begin{array}{c}
 \boxed{\frac{S}{\exists} \diamond \frac{R}{\exists}} \xrightarrow{\text{definition of } \diamond, \text{union} = E(\exists)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\right) E(\exists)} \xrightarrow{E \text{ a functor, } E(X)E(Y) = E(XY)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\exists\right)} \\
 \xrightarrow{(10b), \frac{R}{\exists}\exists = R} \boxed{\frac{S}{\exists} E(R)} \xrightarrow{(10b), \text{absorption}} \boxed{\frac{SR}{\exists}}
 \end{array}$$

the first three steps only unfold $\diamond$ — E is a functor, so $E\left(\frac{R}{\exists}\right)\text{union} = E\left(\frac{R}{\exists}\exists\right) = E(R)$ and the union is gone.

Absorption, the row of (10b), is the whole law, and it is the functoriality of $\frac{1}{\exists}$.

1 goes to $\frac{1}{\exists}$, the Kleisli identity, by definition — so $\frac{1}{\exists}$ is an isomorphism of categories $\mathcal{A} \cong \mathbf{Kleisli}(E)$, and §Section 10.1's $i \dashv E$ is the Kleisli adjunction. i is the identity on objects, so every object of the allegory is free.